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Data Scientist | Machine Learning | Generative AI | RAG | Automotive & Embedded Analytics

Rakesh

PROFILE

Data Scientist with 3+ years of experience in machine learning, deep learning, and end-to-end AI solution development. Specialized in Python, NLP, Generative AI, and RAG-based document intelligence. Skilled in translating business and system requirements into high-impact AI solutions.

SKILLS

- **Programming & Data:** Python, SQL, Pandas, NumPy, EDA, Data Preprocessing
- **Machine Learning:** Regression, Classification, Clustering, Anomaly Detection, Feature Engineering
- **Time Series & Signals:** Time Series Forecasting, Sequence Modeling, Vehicle Signal Analysis
- **Deep Learning & NLP:** CNN, LSTM, NLP, Text Analytics, Embeddings, Semantic Search
- **Generative AI:** LLMs, RAG, Prompt Engineering, Document Intelligence
- **Requirements & Automation:** SWE.6 Requirement Derivation (ASPICE), Auto Test Case Generation, Traceability
- **Automotive & Embedded:** ECU Requirements, CAN, UDS, Diagnostics, HIL/SIL (Exposure)
- **Testing & Automation:** Python Automation, PyTest, Test Data Analysis
- **MLOps & Deployment:** REST APIs, MLflow, Docker (Basics)
- **Visualization & Tools:** Matplotlib, Power BI (Basics), Git, Jupyter,

PROFESSIONAL SUMMARY

- Data Scientist and Applied AI Engineer with hands-on experience across the **complete Data Science lifecycle** — data ingestion, preprocessing, EDA, modelling, validation, and production deployment
- Strong expertise in **Python, Machine Learning, Deep Learning, NLP, and statistical modelling** using modern ML frameworks
- Proven experience in building **predictive models, time-series analysis, anomaly detection, and classification/regression solutions**
- Hands-on experience with **Generative AI and Retrieval-Augmented Generation (RAG)** for requirement analysis, document intelligence, and knowledge extraction
- Developed **AI-based automatic test case generation** and requirement-to-test traceability solutions
- Strong exposure to **automotive and embedded systems data**, including ECU requirements, CAN signals, diagnostics, and test automation
- Experience in analysing **structured, semi-structured, and unstructured data** (CSV, JSON, HTML, PDF)
- Adept at translating **business, system, and customer requirements into scalable, explainable AI solutions**
- Experienced in integrating AI models into **real-world engineering workflows**, improving efficiency, quality, and decision-making
- Proven ability to deliver **production-ready, high-impact AI solutions** aligned with business and engineering goals

EXPERIENCE

Company Name: KPIT, Pune

Designation: AI / Data Scientist – Automotive

Project: **RAG-Based Customer Requirement to SWE.6 Software Unit Requirement Derivation**

Problem Statement:

Automotive customer requirements (OEM-level, high-level, unstructured) need to be **systematically derived into ASPICE-compliant SWE.6 Software Unit Requirements**, which is time-consuming and error-prone.

Solution:

Designed a **Retrieval-Augmented Generation (RAG) system** to automatically derive **SWE.6-level software unit requirements** from customer and system-level requirements.

Approach:

- Ingested **customer requirements, SYS.2, SWE.1–SWE.5 artifacts** (PDF, DOCX, TML)
- Applied **document chunking + embeddings** for semantic retrieval
- Used **LLM with RAG** to:
  - Understand intent, constraints, and safety aspects
  - Derive **low-level, testable SWE.6 requirements**
- Embedded **ASPICE rule-based validation** (shall statements, atomicity, traceability)
- Linked derived SWE.6 requirements back to parent requirements

Technology Stack

Python, Lang Chain, LLMs, FAISS/Chroma, NLP, Automotive SPICE (ASPICE)

Outcome / Impact

- Reduced manual SWE.6 derivation effort by **~65%**
- Improved **requirement consistency and traceability**
- Enabled faster compliance readiness for **ASPICE audits**

PROJECT 2: **AI-Based Automatic Test Case Generation from Automotive Requirements**

Domain: ECU Testing | Test Automation | AI

Role: AI Engineer / Test Automation Specialist

Problem Statement

Manual test case creation from requirements is slow and inconsistent, especially for large automotive ECUs with frequent requirement changes.

Solution

Built an AI-driven test case generation framework that converts textual requirements into structured test cases automatically.

Approach

- Parsed requirements from DOORS / JSON / CSV
- Applied RAG + LLM to extract:
  - Preconditions
  - Test steps
  - Expected results
- Auto-generated:
  - System test cases
  - Software test cases

Mapped test cases to CAN signals, UDS services, and ECU states

Exported test cases to automation-ready Python framework

Technology Stack

- Python, LLMs, RAG, Automotive Diagnostics (CAN, UDS)

Outcome / Impact

- Reduced test design time by 70%
- Enabled direct integration with HIL/SIL automation
- Enabled direct integration with HIL/SIL automation

PROJECT 3: ADAS

- **Project Description:**

The project involved end-to-end validation of **ADAS and Driver Assistance Technologies (DAT)** for Ford's next-generation vehicles in the North American market. Features under test include **Pre-Collision Assist**, **Traffic Jam Assist**, **Adaptive Cruise Control**, **Rear Cross Traffic Alert**, **Automatic Emergency Braking**, and **Lane Centering/Assist**. Additionally, **End-of- Line (EOL) diagnostics** such as **Loss of Communication DTCs** and **Invalid Data Flag detection** were thoroughly verified

Role and Responsibilities:

- **Software Validation in Simulated HIL Environment:**
- Performed functional validation of ECU software using **Vector VT System** with integrated plant models.
- Conducted **software flashing** to the ECU and configured simulation setups for test execution.
- Visualized and analyzed ECU runtime parameters and sensor/actuator signals using **Canapé and Canoe**.
- **Test Coverage and Execution:**
- Conducted comprehensive validation across **Communication**, **Diagnostics (UDS)**, and **Failsafe**
- Test categories.
- Executed both **manual and automated test cases**, ensuring feature robustness under various simulated driving conditions.
- **Python-Based Automation Framework:**
- Developed and integrated a **Python automation framework** to streamline test execution, result logging, and report generation.
- Scripted reusable Python modules for diagnostic service automation and CAN signal manipulation.
- Achieved a significant reduction in manual effort and increased regression testing efficiency.

EDUCATION **Bachelor of Technology in Electrical and Electronics Engineering from SIDDHARTHA INSTITUTE OF ENGINEERING AND TECHNOLOGY, JNTUH with the aggregate of 66%.**

Personal Details

Father Name: SRIHARI CHUNCHU

Languages: English

(Rakesh)